AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2014

General Mathematics Paper II

Time: 2 hours 20 minutes Marks: 45

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are TEN questions. Answer ALL questions. Choices are specified inside the paper

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use colour pencils.
   DO NOT use staples, paper clips, glue or correcting fluid.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 3 Marks)

Blue and yellow paints are mixed in the ratio 5:4 to make a colour of Ambreen's choice. If 18 litres of paint is required to paint the exterior of her apartment, then calculate the total cost of paint by using the given table.

Paint Colour	Price (per litre)
Blue	Rs. 800
Yellow	Rs. 820

(ATTEMPT EITHER PART a OR PART b OF Q.2.)

Q.2. (Total 4 Marks)

a. Mr. Ahmed saves Rs. 2,000,000 annually from his business and Rs. 25,000 monthly from the house rent he receives. Find the amount of zakat he must pay; the zakat rate being 2.5 %.

b. Mr. Salman owns a house, a car and a shop worth Rs. 4,000,000, Rs. 600,000 and Rs. 3,500,000 respectively. He wants to divide his property among his legal heirs/heiresses. His wife receives $\frac{1}{8}th$ of the property while his mother receives $\frac{1}{6}th$ of the property. The remaining property is divided between his son and daughter in such a way that the son gets twice the share of his daughter. Find the share inherited by his mother, wife and daughter.

(ATTEMPT EITHER PART a OR PART b OF Q.3.)

Q.3. (Total 5 Marks)

a. Mr. Shayan is planning to buy a car in June. The price of the car was Rs. 1,200,000 in January. The price was increased in February by 10% and then increased by 15% in May. What will Mr. Shayan have to pay for the car on 1st June? Find the difference of the car's price in January to its price in June.

b. Three partners Fozia, Arif and Kamran invested $ 1,000,000 in a business. They invested in a ratio 5:7:8 respectively. Find the share of each partner in the investment. If they earned a profit of $ 30, 000, then find the share of each partner in the profit.

Q.4. (Total 3 Marks)

Farzana invested Rs. 150,000 at a simple interest rate of 12% per annum. What will be her accumulated amount after 5 years? How many years will she take to double her interest assuming that the interest rate remains constant?

Q.5. (Total 5 Marks)

A person's salary is Rs. 60, 000 per month. Before deduction of income tax the following monthly deductions are made from his salary

- Provident fund Rs. 6,000
- Professional fee Rs. 2,000
- EOBI charges Rs. 100
- Miscellaneous charges Rs. 900

Find his annual taxable income and his annual payable tax if the rate of tax is 9% per annum. Also find his annual income after deduction of tax.

(ATTEMPT EITHER PART a OR PART b OF Q.6.)

Q.6. (Total 5 Marks)

a. Simplify $\sqrt[3]{\frac{64a^2b^6c^4}{8a^{-10}b^3c}}$

(ATTEMPT EITHER PART a OR PART b OF Q.6.)

b. Do as directed.

i. Convert $2^x = 3$ to logarithmic form. (1 Mark)

ii. Find the characteristic of log 2305 (1 Mark)

iii. Find the value of log 10000 (1 Mark)

iv. Find x if $\log_5 x = 1$ (1 Mark)

v. Write $\log x - 3\log y$ as a single logarithm. (1 Mark)

Q.7. (Total 4 Marks)

Find FOUR arithmetic means between 2 and 8

(ATTEMPT EITHER PART a OR PART b OF Q.8.)

Q.8. (Total 5 Marks)

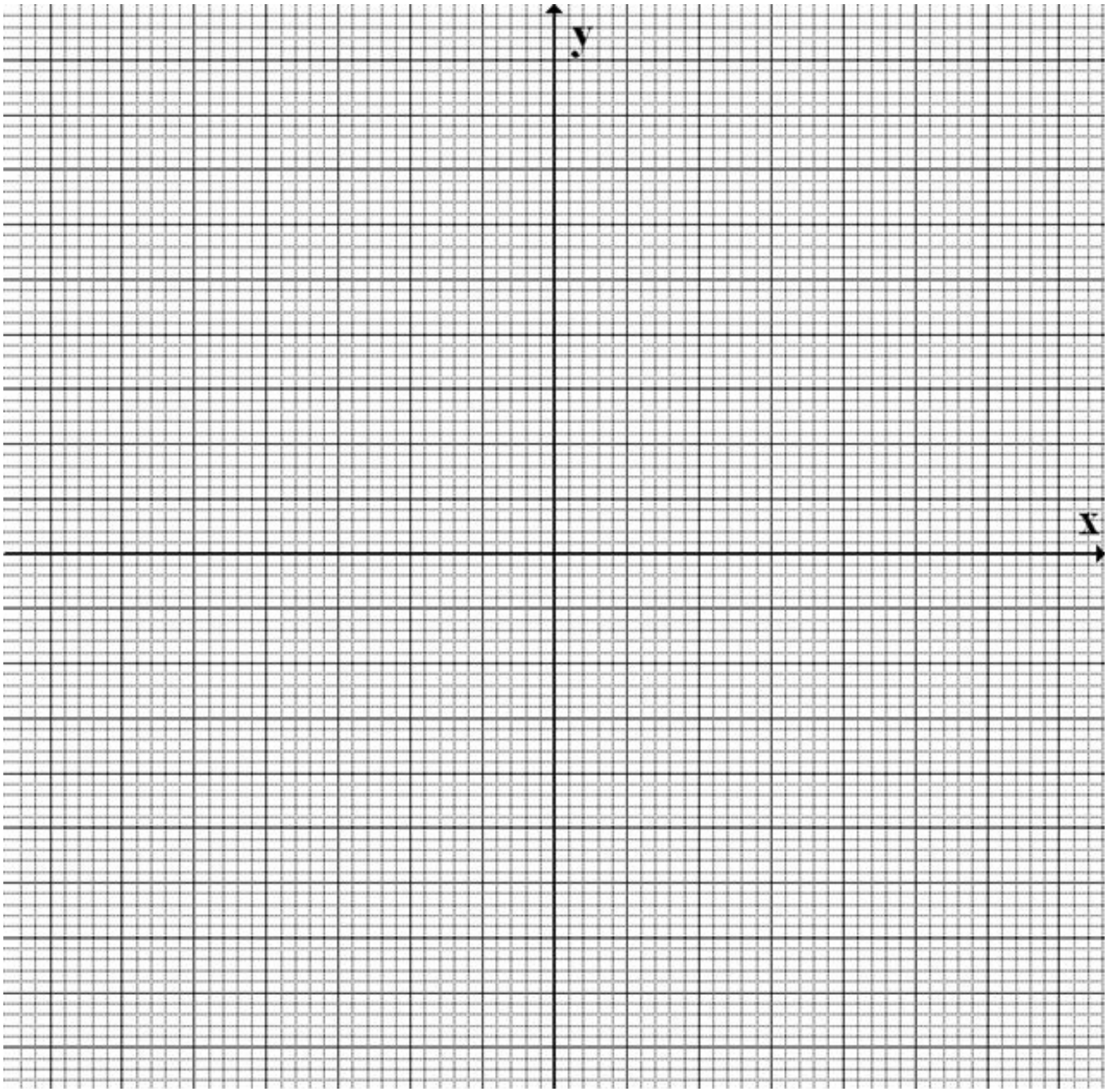
a. For sets $U = \{2, 4, 6, 8, 10, 12\}$, $A = \{2, 8, 12\}$ and $B = \{2, 4, 6, 8\}$, find $A \cup B$, $A \cap B$, $A - B$, A^c and B^c.

b. For set $A = \{x, y, z\}$, find $A \times A$, an onto function, an into function and a one-one function.

Q.9. (Total 5 Marks)

For the equation $y = 3x - 2$ fill in the following table and plot the graph.

x				
y				

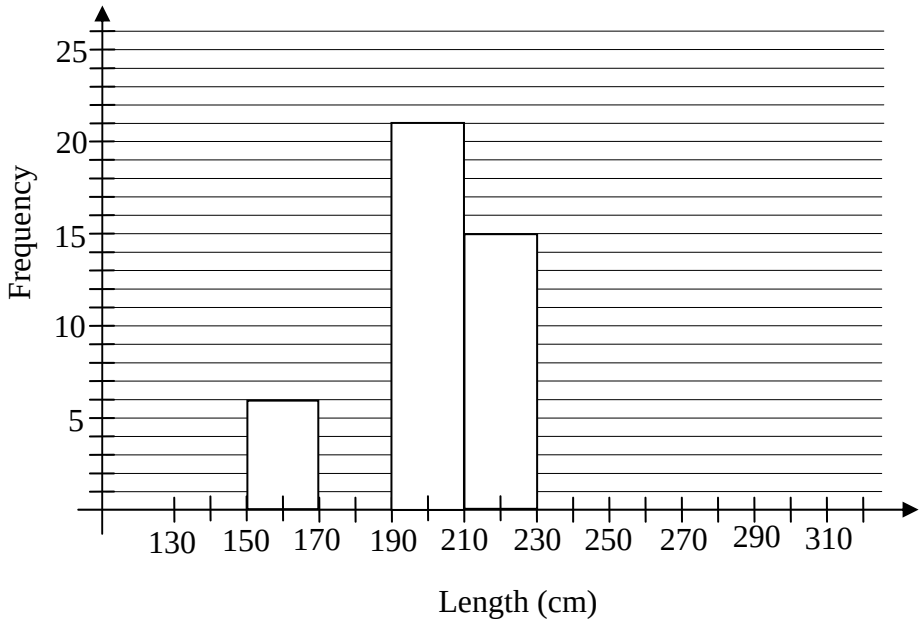


(ATTEMPT EITHER PART a OR PART b OF Q.10.)

Q.10. (Total 6 Marks)

a. The table and histogram below are both incomplete. They represent the information about the length of sugarcanes after cutting the crop. Complete both the table and the histogram.

Length (cm)	150-170	170-190	190-210	210-230	230-250
Frequency		10	21		6



i. Find the total number of sugarcanes taken as the sample.

ii. How many sugarcanes have lengths between 170 cm and 230 cm?

(ATTEMPT EITHER PART a OR PART b OF Q.10.)

b. Mr. Nadeem is an incharge of an internet club in a school. He has recorded the time (in minutes) spent on the internet by each of 12 students in one day. Below are the results:

68	79	91	77	88	82
76	65	82	68	77	86

Find the range, standard deviation and variance of the data. (6 Marks)

END OF PAPER